PHY 211: ELEMENTARY MODERN PHYSICS
Module 3: Week 9 - BLACK BODY RADIATION

Dr JOLAYEMI O. R.

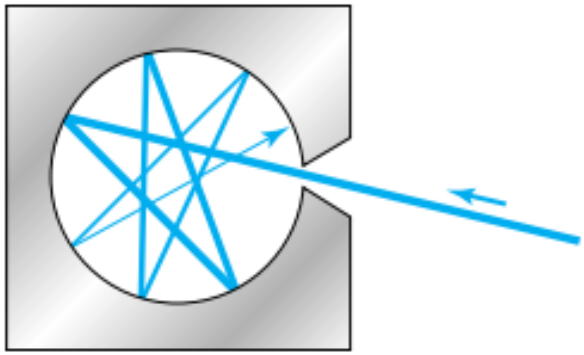
Department of Physics
College of Science & Technology
COVENANT UNIVERSITY

2025/2026 Session

Introduction

- When matter is heated, it emits radiation. We can determine experimentally that a broad spectrum of wavelengths is emitted when matter is heated. This process was of great interest to physicists of the nineteenth century. They measured the intensity of radiation being emitted as a function of material, temperature, and wavelength.
- All bodies simultaneously emit and absorb radiation. When a body's temperature is constant in time, the body is said to be in thermal equilibrium with its surroundings. In order for the temperature to be constant, the body must absorb thermal energy at the same rate as it emits it. This implies that a good thermal emitter is also a good absorber.
- For thermal radiation the simplest case is a **blackbody**, which has the ideal property that it absorbs all the radiation falling on it and reflects none.

Blackbody Radiation

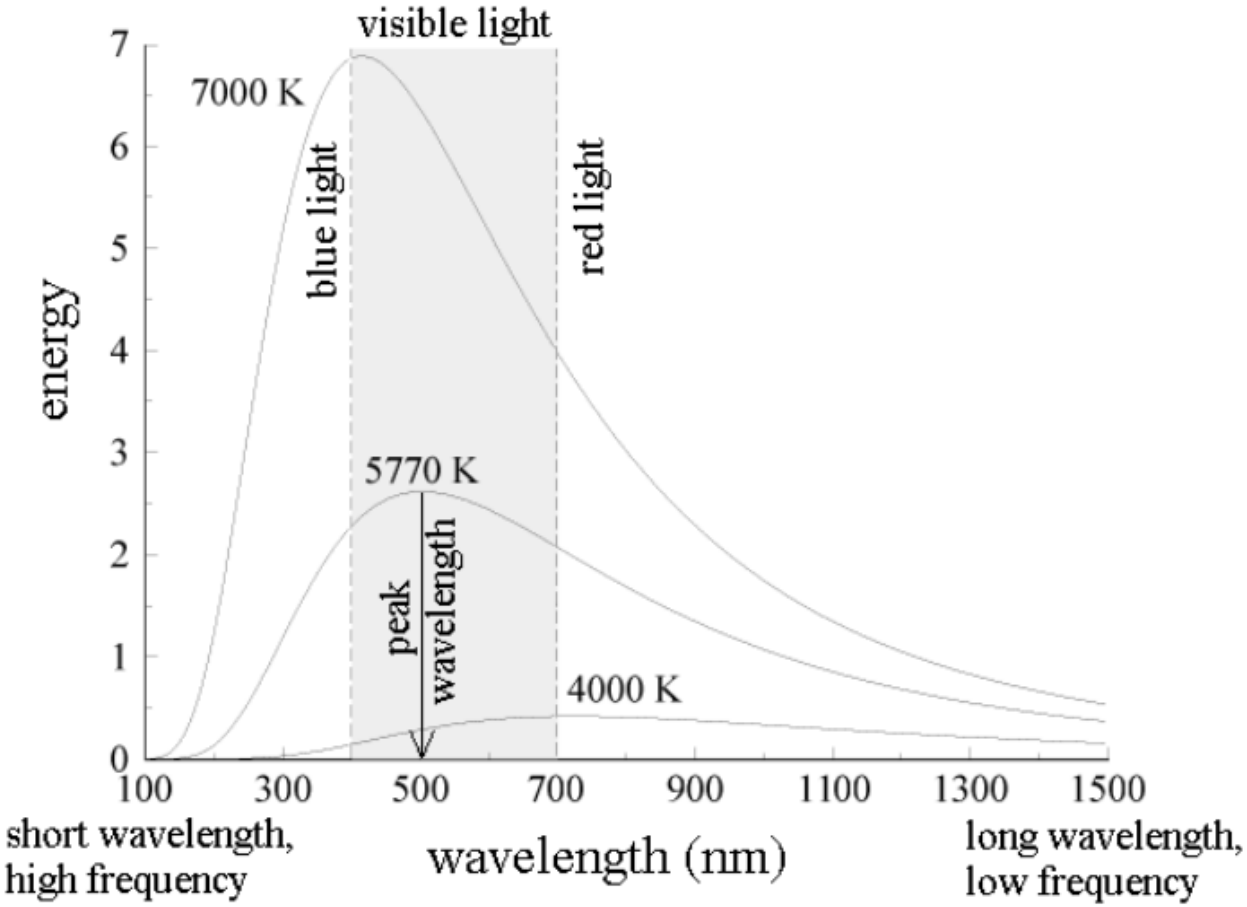


The simplest way to construct a blackbody is to drill a small hole in the wall of a hollow container. Radiation entering the hole will be reflected around inside the container and then eventually absorbed. Only a small fraction of the entering rays will be reemitted through the hole. If the blackbody is in thermal equilibrium, then it must also be an excellent emitter of radiation.

Blackbody radiation is theoretically interesting because of its universal character: **the radiation properties of the blackbody (that is, the cavity) are independent of the particular material of which the container is made**. Physicist can study the properties of blackbody radiation. The intensity versus wavelength (called spectral distribution) at fixed temperatures of the radaition can be studied without having to understand the details of emission or absorption by a particular kind of atom.

NOTE: The thermal radiation of a black body is electromagnetic radiation.

Spectral distribution of radiation emitted from a blackbody for different blackbody temperatures



The intensity $\ell(\lambda, T)$ is the total power radiated per unit area per unit wavelength at a given temperature.

(1): Observation from the Blackbody spectrum

The first main features of the spectral intensity of a Blackbody emitter is as follows:

- The wavelength of peak emission shifts to shorter (bluer) wavelengths as the temperature of the Blackbody increases.

This observation is expressed in the **Wien's displacement law** and takes the form:

$$\lambda_{max} T = 2.898 \times 10^{-3} mK \tag{1}$$

where λ_{max} is the wavelength of the peak of the spectral distribution at a given temperature.

Even on its own this is a very useful law: One can observe the spectral intensity of the Sun and infer the temperature of the photosphere; by observing the spectrum of the Cosmic Microwave Background we can measure the "temperature" of the universe; you could even observe the emission from an iron bar and work out whether it was safe to pick up!

(2): Observation from the Blackbody spectrum

The second main features of the spectral intensity of a Blackbody emitter is as follows:

- The total power radiated increases with the temperature.

We can quantify the second observation by integrating the quantity $\ell(\lambda, T)$ over all wavelengths to find the power per unit area at temperature T

$$R(T) = \int_0^\infty \ell(\lambda, T) d\lambda \tag{2}$$

Josef Stefan found empirically in 1879, and Boltzmann demonstrated theoretically several years later, that $R(T)$ is related to the temperature by

$$R(T) = \epsilon \sigma T^4 \tag{3}$$

This is known as the **Stefan-Boltzmann law**, with the constant σ experimentally measured to be $5.6705 \times 10^{-8} Wm^{-2}k^{-4}$.

The Stefan-Boltzmann law equation can be applied to any material for which the emissivity is known. The emissivity ϵ ($\epsilon = 1$ for an idealized blackbody) is simply the ratio of the emissive power of an object to that of an ideal blackbody and is always less than 1. Thus, Equation (3) is a useful and valuable relation for practical scientific and and engineering work.

Class Exercise

A furnace has walls of temperature 1600^0C. What is the wavelength of maximum intensity emitted when a small door is opened?

Class Exercise

A furnace has walls of temperature 1600^0C. What is the wavelength of maximum intensity emitted when a small door is opened?

SOLUTION

We assume the furnace with a small door open is a blackbody so that we can determine λ_{max} from the Wien's Displacement Law

$$\lambda_{max} T = 2.898 \times 10^{-3} mK$$

We first convert the temperature to kelvin.

$$T = (1600 + 273)K = 1873K$$
$$\lambda_{max}(1873K) = 2.898 \times 10^{-3} m.K$$
$$\therefore \lambda_{max} = 1.55 \times 10^{-6} m$$

Class Exercise

The wavelength of maximum intensity of the sun's radiation is observed to be near $500nm$. Assume the sun to be a blackbody and calculate the sun's surface temperature.

Class Exercise

The wavelength of maximum intensity of the sun's radiation is observed to be near $500nm$. Assume the sun to be a blackbody and calculate the sun's surface temperature.

SOLUTION

From the Wien's Displacement Law, we can calculate the sun's surface temperature with $\lambda_{max} = 500nm$

$$\lambda_{max} T_{sun} = 2.898 \times 10^{-3} m.K$$

$$(500nm) T_{sun} = 2.898 \times 10^{-3} m.K \frac{10^9 nm}{m}$$

$$T_{sun} = \frac{2.898 \times 10^6}{500} K = 5800K$$

Lord Rayleigh's Classical Theory and Blackbody Spectral Distribution Discrepancy



The Lord Rayleigh
1842 AD - 1919 AD
AKA John William Strutt, 3rd Baron Rayleigh

The nature of the blackbody spectrum can be understood from classical electromagnetic theory, together with statistical thermodynamics. The radiation emitted from a blackbody can be expressed as a superposition of electromagnetic waves of different frequencies within the cavity. That is, radiation of a given frequency is represented by a standing wave inside the cavity. The equipartition theorem of thermodynamics assigns equal average energy kT to each possible wave configuration.

Lord Rayleigh used the classical theories of electromagnetism and thermodynamics to show in June 1900 that the blackbody spectral distribution should have a $\frac{1}{\lambda^4}$ dependence, **which is completely inconsistent with the experimental result at low wavelength.**

Rayleigh-Jeans formula

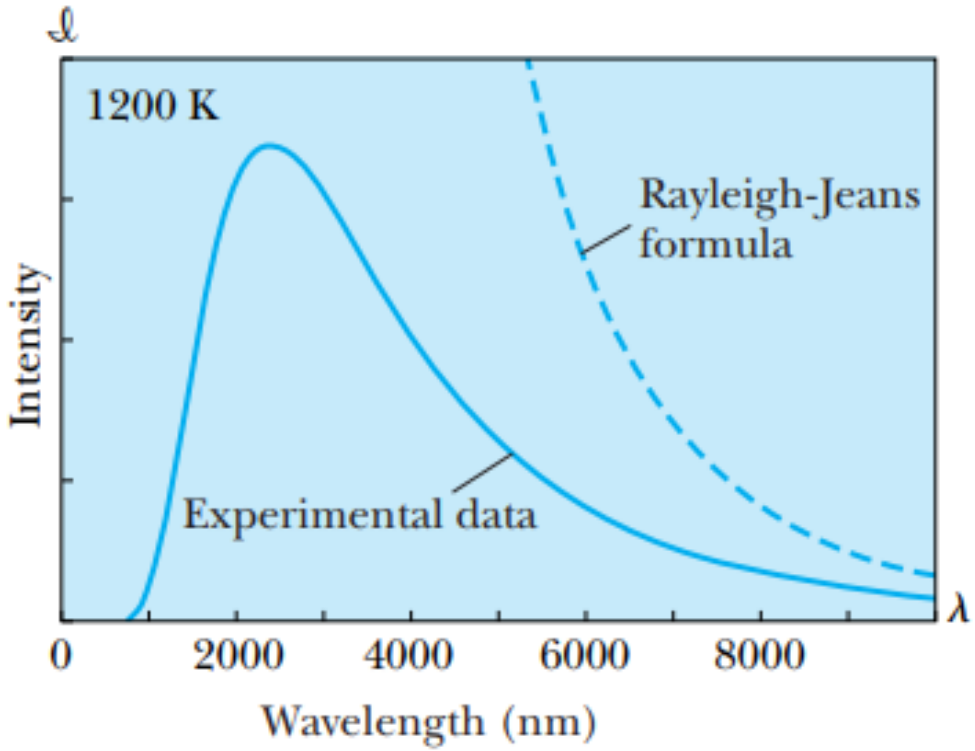


In 1905, Sir James Jeans helped Rayleigh determine the factor in front of this distribution, they presented their complete result to be:

$$\ell(\lambda, T) = \frac{2\pi ckT}{\lambda^4} \tag{4}$$

This result i.e. Equation (4) is known as the Rayleigh-Jeans formula, and it is the best formulation that classical theory can provide to describe blackbody radiation.

Rayleigh-Jeans result compared with experimental data



The spectral distribution calculated by the Rayleigh-Jeans formula is compared with blackbody radiation experimental data at $1200K$. The formula approaches the data at large wavelengths but disagrees badly at low wavelength.

Theoretical attempts to explain the spectrum of a Black-body

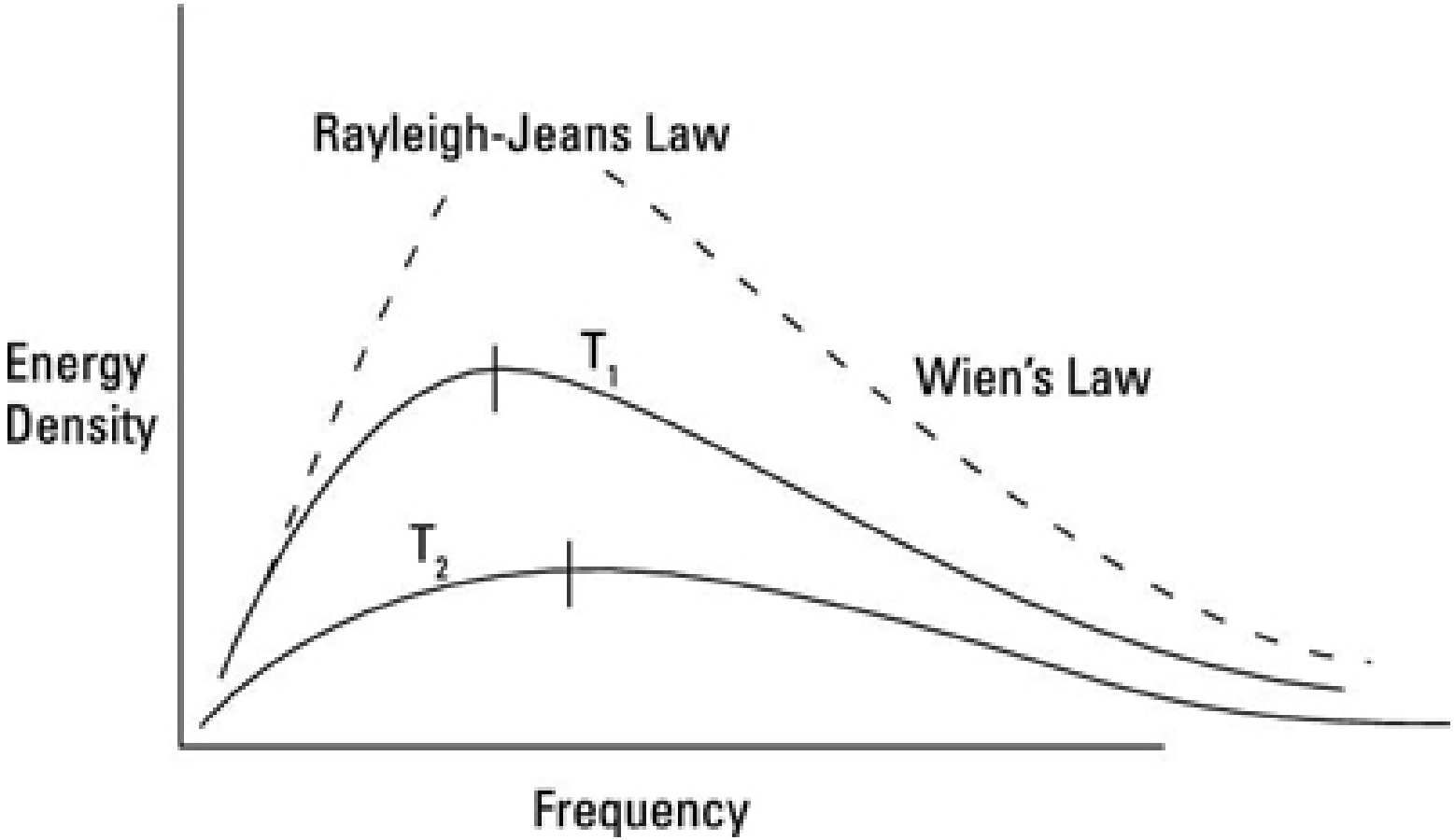
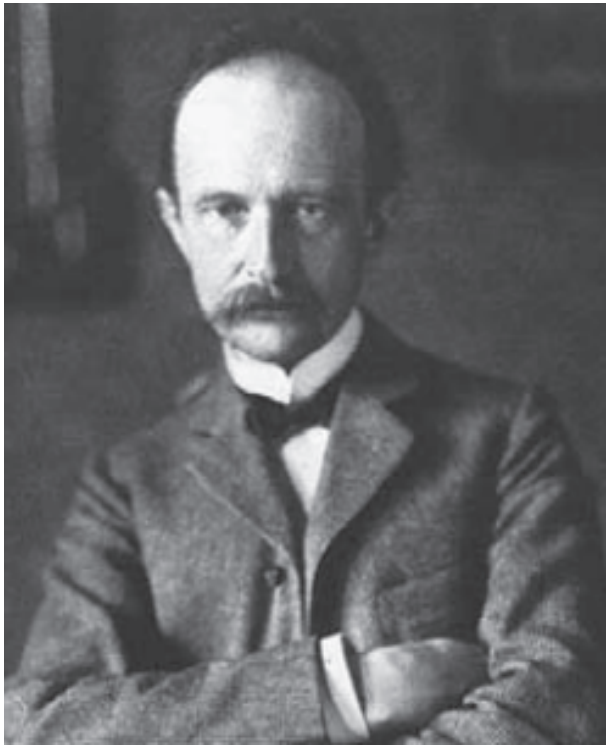


Figure: Black-body radiation spectrum

Max Planck



- The German Max Planck was an expert on the second law of thermodynamics. He set out in 1895 to examine the irreversibility of radiation processes.
- Planck tried various functions of wavelength and temperature until he found a single formula that fit the measurements of $\ell(\lambda, T))$ over the entire wavelength range.
- Planck was simply looking for a formula that fit the known blackbody spectral distribution. Planck reported his formula in October 1900, but he realized a month later it was nothing but an inspired guess.

Planck's Radiation Law

Planck followed Hertz's work using oscillators to confirm the existence of Maxwell's electromagnetic waves, and lacking detailed information about the atomic composition of the cavity walls, Planck assumed that the radiation in the cavity was emitted (and absorbed) by some sort of "oscillators" that were contained in the walls. When adding up the energies of the oscillators, he assumed (for convenience) that each one had an energy that was an integral multiple of hf, where f is the frequency of the oscillating wave and h is a constant. He was applying a technique invented by Boltzmann, and Planck ultimately expected to take the limit $h \to 0$, to include all the possibilities. However, he noticed that by keeping h nonzero, he arrived at the equation needed for $\ell(\lambda, T)$:

$$\ell(\lambda, T) = \frac{2\pi c^2 h}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda k T}} - 1} \tag{5}$$

Equation (5) is Planck's radiation law, which he reported in December 1900.

Planck's Modifications to Classical Theory for Agreement with Experimental Data

Planck with the experimental data only made two important modifications of classical theory:

- The oscillators (of electromagnetic origin) can only have certain discrete energies determined by $E_n = nhf$, where n is an integer, f is the frequency, and h is called Planck's constant and has the value:

$$h = 6.6261 \times 10^{-34} J.s$$

- The oscillators can absorb or emit energy in discrete multiples of the fundamental quantum of energy given by:

$$\Delta E = hf$$

Planck found these results quite disturbing and spent several years trying to find a way to keep the agreement with experiment while letting $h \to 0$. Each attempt failed, and Planck's quantum result became one of the cornerstones of modern science.

Questions



1. Show that Wien's displacement law follows from Planck's radiation law.
2. Use Planck's radiation law to find the Stefan-Boltzmann law.
3. Show that the Planck radiation law agrees with the Rayleigh-Jeans formula for large wavelengths.
4. Show that Planck's radiation law resolves the *ultraviolet catastrophe*.

APPENDIX

Show that Wien's displacement law follows from Planck's radiation law.

Wien's law, Equation (3.14), refers to the wavelength for which $\mathcal{L}(\lambda, T)$ is a maximum for a given temperature. From calculus we know we can find the maximum value of a function for a certain parameter by taking the derivative of the function with respect to the parameter, set the derivative to zero, and solve for the parameter.

Solution

Therefore, to find the value of the Planck radiation law for a given wavelength we set $d\mathcal{L}/d\lambda = 0$ and solve for λ.

$$\frac{d\mathcal{L}(\lambda, T)}{d\lambda} = 0 \quad \text{for } \lambda = \lambda_{\max}$$

$$2\pi c^2 h \frac{d}{d\lambda}[\lambda^{-5}(e^{hc/\lambda kT} - 1)^{-1}]\bigg|_{\lambda_{\max}} = 0$$

$$-5\lambda_{\max}^{-6}(e^{hc/\lambda_{\max}kT} - 1)^{-1} - \lambda_{\max}^{-5}(e^{hc/\lambda_{\max}kT} - 1)^{-2} \times \left(\frac{-hc}{kT\lambda_{\max}^2}\right)e^{hc/\lambda_{\max}kT} = 0$$

Multiplying by $\lambda_{\max}^6(e^{hc/\lambda_{\max}kT} - 1)$ results in

$$-5 + \frac{hc}{\lambda_{\max}kT}\left(\frac{e^{hc/\lambda_{\max}kT}}{e^{hc/\lambda_{\max}kT} - 1}\right) = 0$$

Let

$$x = \frac{hc}{\lambda_{\max}kT}$$

Then

$$-5 + \frac{xe^x}{e^x - 1} = 0$$

and

$$xe^x = 5(e^x - 1)$$

This is a transcendental equation and can be solved numerically (try it!) with the result $x \approx 4.966$, and therefore

$$\frac{hc}{\lambda_{\max}kT} = 4.966$$

$$\lambda_{\max}T = \frac{hc}{4.966\,k} = \frac{1240\ \text{eV}\cdot\text{nm}}{4.966\left(8.617 \times 10^{-5}\dfrac{\text{eV}}{\text{K}}\right)}\frac{10^{-9}\ \text{m}}{\text{nm}}$$

and finally,

$$\lambda_{\max}T = 2.898 \times 10^{-3}\ \text{m}\cdot\text{K}$$

which is the empirically determined Wien's displacement law.